



NPTEL Online Certification Course
Applied Accelerated Artificial Intelligence
Assignment 1: Detailed Solution
Indian Institute of Technology Guwahati

Question 1:

What is Memory Wall?

- A) The high latency difference between DRAM & CPU
- B) The lack of storage space for large files in secondary hard drives
- C) The physical limitation of the silicon die size in processors
- D) Speed of L1 cache is around 4 cycles per core/private

Correct Answer: A) The high latency difference between DRAM & CPU

Detailed Solution: The "Memory Wall" refers to the growing performance gap between how fast a CPU can execute instructions and how fast data can be transferred from main memory (DRAM). While processor speeds have increased exponentially over decades, memory access times have improved at a much slower rate, causing the CPU to frequently sit idle while waiting for data.

Question 2

What is Re-Order buffer?

- A) Stores recently used pages
- B) Looks ahead for dependent & independent instructions in advance
- C) Stores the recently used instructions
- D) Used in In-Order Execution

Correct Answer: B) Looks ahead for dependent & independent instructions in advance

Detailed Solution: The Re-Order Buffer (ROB) is a critical component used in Out-of-Order (OoO) execution processors. It tracks instructions that are executed out of chronological order due to structural or data dependencies, allowing the processor to safely check ahead for independent instructions while ensuring they are committed (retired) in the correct, original program order.

Question 3

For data level parallelism, which type of extensions are used in Modern systems?

- A) SIMD
- B) SISD
- C) MISD
- D) MIMD

Correct Answer: A) SIMD

Detailed Solution: SIMD stands for Single Instruction, Multiple Data. It allows a processor architecture to apply a single machine instruction across a vector of multiple data points simultaneously. Modern examples include Intel AVX/AVX-512 and ARM Neon, which are fundamental for accelerating vector/matrix mathematics.

Question 4

The left & right side of intersection of peak performance curve & peak memory bandwidth curve represents?

- A) Memory bound & compute intensive workload
- B) Compute intensive & memory bound workload
- C) Cannot differentiate
- D) None of the above

Correct Answer: B) Compute intensive & memory bound workload

Detailed Solution: This question refers to the Roofline Performance Model. The point of intersection represents the hardware's machine balance. To the left of this inflection point, the workload's performance is limited by the memory bandwidth (Memory-bound). To the right of this point, the performance hits the hardware plateau and is limited by raw processing speed (Compute-bound / Compute-intensive).

Question 5

CPU cannot do transformation attention (large matrix multiplication). Why?

- A) Because it does not fall under the memory roof for CPUs only.
- B) The statement is false, it can be done by CPU.
- C) Because it is above the maximum compute performance of CPU
- D) It is neither done by CPU not by GPU

Correct Answer: C) Because it is above the maximum compute performance of CPU

Detailed Solution: Large matrix operations like Transformer Attention mechanisms require trillions of calculations per second (FLOPs). While a CPU is *technically* capable of running the code logic, its physical architectural limits (fewer cores, lack of specialized tensor processing units) mean these operations require compute performance levels that vastly exceed what a standard CPU can physically deliver in a reasonable timeframe.

Question 6

LLM inference & DNN training are:

- A) Memory Bound & Compute Bound
- B) Compute Bound & Memory Bound
- C) Memory Bound & Memory Bound
- D) Compute Bound & Compute Bound

Correct Answer: A) Memory Bound & Compute Bound

Detailed Solution: 11

LLM Inference is primarily Memory-bound because it processes tokens sequentially (batch size is often small), requiring the entire set of model weights to be fetched from memory for every single token generated.

DNN Training is primarily Compute-bound because it processes large batch sizes over complex forward and backward passes, maximizing data reuse inside processor registers/caches.

Question 7

DDR5 Memory have:

- A) 2500 - 6400 MT/s, 32 bit wide bus
- B) 4800 - 6400 MT/s, 32 bit wide bus
- C) 1200 - 4800 MT/s, 64 bit wide bus
- D) 4800 - 6400 MT/s, 64 bit wide bus

Correct Answer: D) 4800 - 6400 MT/s, 64 bit wide bus

Detailed Solution: Standard JEDEC DDR5 memory modules operate within baseline speeds of 4800 to 6400 MT/s (MegaTransfers per second). A standard single-channel desktop/server DDR5 DIMM maintains a total data bus width of 64 bits (architecturally split into two independent 32-bit subchannels to improve efficiency).

Question 8

ECC (Error correcting code) Memory in AI servers do:

- A) Detect and correct single bit and double bit errors
- B) Detect single bit errors, correct double bit errors
- C) Detect double bit errors, correct single bit errors
- D) Correct both single bit and double bit errors

Correct Answer: C) Detect double bit errors, correct single bit errors

Detailed Solution: Standard enterprise ECC memory uses SECDED (Single Error Correction,

Double Error Detection) algorithms. It can seamlessly identify and correct a single-bit corruption on the fly without system interruption, and safely detect (but not fix) a double-bit memory error, triggering a system alert to prevent data corruption.

Question 9

In NUMA architecture (like DGX H100), accessing cross socket memory adds around:

- A) 2x latency
- B) 3x latency
- C) 4x latency
- D) 6x latency

Correct Answer: A) 2x latency

Detailed Solution: In Non-Uniform Memory Access (NUMA) systems, a CPU or GPU has "local" memory directly wired to its socket. If a processor must access memory wired to a *different* socket (cross-socket communication), the data must traverse interconnect fabrics (like UPI or Infinity Fabric), which adds an overhead penalty of approximately 2x latency compared to a direct local fetch.

Question 10

Tick the correct option regarding warps in CUDA:

- A) Warps is 32 threads executing same instruction on multiple data
- B) Warps is 64 threads executing different instruction on multiple data
- C) Warps is 32 threads executing same instruction on single data
- D) Warps is 64 threads executing same instruction on multiple data

Correct Answer: A) Warps is 32 threads executing same instruction on multiple data

Detailed Solution: In NVIDIA's CUDA programming model, a "Warp" is the basic unit of execution. It consists of a group of 32 threads that are executed concurrently in a SIMT (Single Instruction, Multiple Threads) fashion, meaning all 32 threads execute the exact same instruction at any given clock cycle across different pieces of data.

Question 11

The precision in Tensor cores is _____ to TFLOPS.

- A) Exponentially increasing
- B) Exponentially decreasing
- C) Inversely proportional
- D) Directly proportional

Correct Answer: C) Inversely proportional

Detailed Solution: As numerical precision decreases (e.g., shifting down from FP32 to FP16, BF16, or INT8), the processing hardware can handle far more calculations simultaneously within the same physical silicon footprint. Therefore, lower numerical precision leads to higher raw processing throughput (TFLOPS), making precision and performance throughput inversely proportional.

Question 12

" torch.cuda.amp.autocast() " is used to:

- A) Default to higher precision units
- B) Default to lower precision units
- C) Automatic Mixed Precision
- D) None of the above

Correct Answer: C) Automatic Mixed Precision

Detailed Solution: `torch.cuda.amp.autocast()` is PyTorch's utility handler for Automatic Mixed Precision (AMP) training. It automatically matches specific operations (like matrix multiplications) to faster, lower-precision data types (like float16/bfloat16) while keeping critical operations (like loss scaling and reductions) in float32 to preserve model convergence.

Question 13

Which type of advantage does the HBM3 have over H100 & AMD MI300X?

- A) Computation advantage
- B) Memory storage advantage
- C) Multi core CPU advantage
- D) Memory Bandwidth advantage

Correct Answer: D) Memory Bandwidth advantage

Detailed Solution: High Bandwidth Memory 3 (HBM3) utilizes a 3D-stacked DRAM design connected via an ultra-wide silicon interposer interface. This provides massive memory bandwidth advantages (reaching terabytes per second) compared to traditional GDDR or DDR memory subsystems, keeping hyper-parallel computing engines properly saturated with data.

Question 14

What is the primary architectural bottleneck that NVIDIA's NVSwitch is designed to resolve?

- A) The storage read/write limitations between local NVMe drives and system memory.
- B) The bandwidth and scalability limits of point-to-point NVLink connections across multiple GPUs.
- C) The latency delays caused by PCIe slot power throttling during deep learning workloads.

- D) The network congestion between different server racks inside a data center over standard Ethernet.

Correct Answer: B) The bandwidth and scalability limits of point-to-point NVLink connections across multiple GPUs.

Detailed Solution: While NVLink allows direct high-speed point-to-point connections between a limited number of local GPUs, scaling beyond a single baseboard requires a centralized crossbar system. NVSwitch enables multiple NVLink links to map across a unified switching fabric, allowing all GPUs inside an enterprise node (or across multiple nodes) to communicate at full NVLink speeds simultaneously.

Question 15

Which of the following functions is NOT a CUDA Runtime API function?

- A) `cudaFree`
- B) `cudaMemcpy()`
- C) `cudaMalloc()`
- D) `cuMemAlloc()`

Correct Answer: D) `cuMemAlloc()`

Detailed Solution: Functions prefixing with lower-case `cuda` (such as `cudaMalloc`, `cudaFree`, `cudaMemcpy`) belong to the higher-level CUDA Runtime API. Functions prefixing with `cu` and utilizing alternative casing (such as `cuMemAlloc`) belong directly to the lower-level CUDA Driver API, which interacts closer to the hardware level.